Agentic Data Pipeline Orchestration for NASA Reanalyses — Colorado Snowpack

MERRA-2 and ERA5, Water Years 1981–2026

Built with coding agents and agent skills

Responsible Gen-AI for NASA Earthdata Hackweek

responsible-genai.hackweek.io

August 28, 2026

Project Planning

- Research ✓ — `research/` — 3 product notes
- Plan ✓ — `plan/` — 2 specifications
- Implement ✓ — `src/` — 26 modules
- Verify — `tests/` — 329 tests

The Plan Fixes

- the question
- domain and period
- variables, units, conversions
- the weighting rule
- what is out of scope

And What May Be Claimed

- rank, never magnitude
- every number names its product
- peak SWE needs daily data
- MERRA-2’s 1 April value is degenerate
- say where ERA5T is load-bearing

Agent Skills

`SKILL.md`

`name:`
`description:` *when to use it*

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the traps
the conventions
what may be claimed

the agent matches on the description

. . . so it loads itself,
before it can get it wrong

Science

- `reanalysis-snowpack-comparison`
- `snow-hydrology-fsca-evaluation`
- `modis-merra-regridding`
- `snow-bias-statistics-and-figures`

Run Integrity

- `earthdata-streaming-checkpoints`
- `resumable-fetch-run-integrity`
- `derived-quantity-integrity`

Communication

- `designing-thematic-figures`

Eight skills, written during the work, from what went wrong in it.

The Data Pipeline

MERRA-2	Cloud OPeNDAP • `earthaccess`
ERA5	ARCO-ERA5 Zarr • Google Cloud
MODSCAG	NSIDC DAAC • Earthdata Login
SNOTEL	NRCS AWDB REST • `WTEQ`
3DEP	USGS elevation • staged once

daily checkpoints → water-year statistics → figures and findings

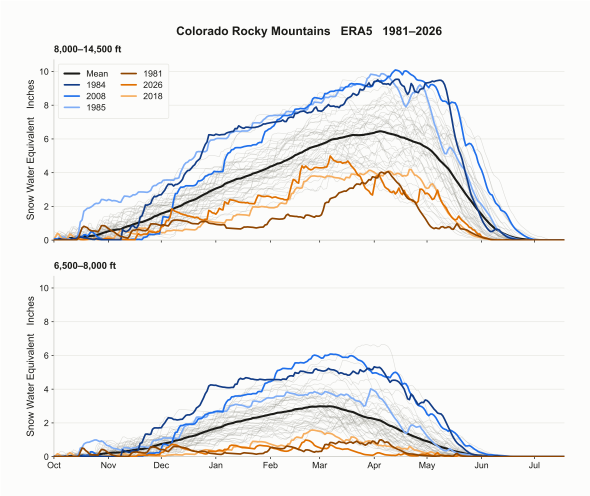
resumable, and the whole report rebuilds from the checkpoints with no network

One Silent Trap per Archive

- **ERA5** `sd` is snow *water equivalent*, not depth. Geometric depth is `sd` / `rsn`
- **MERRA-2** `SNODP` is depth *within* the snow-covered fraction; the grid mean is `SNODP` × `FRSNO`
- **ARCO-ERA5** the time axis runs to 2050. Absent chunks read back as NaN, with no error
- **SNOTEL** every station sits above the domain's median cell. Timing only, never magnitude

Each returns a plausible wrong number rather than an error. Each is now written down in a skill.

ERA5 by Elevation Band



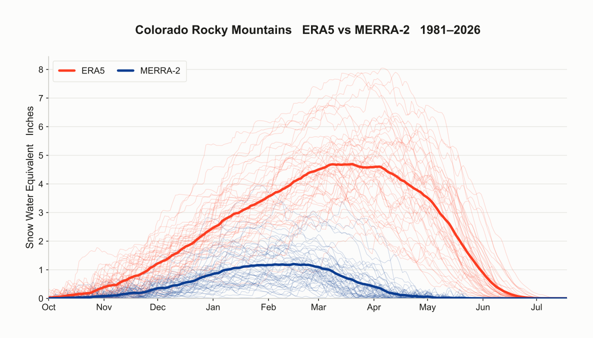
1 April SWE, per cent of band mean

	2023	2026
8,000–14,500 ft	140%	51%
6,500–8,000 ft	297%	5%

WY2026 is the lowest 1 April SWE of the 46 years, at 41% of the record mean. The deficit is not uniform: the lower band all but vanished.

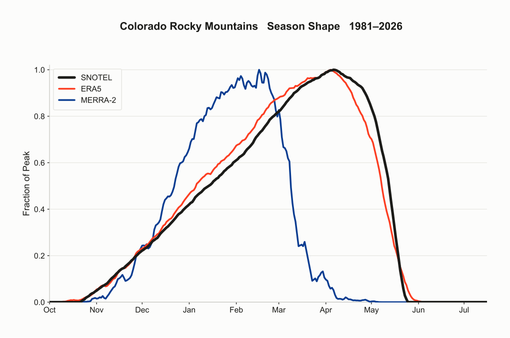
ERA5. The two models disagree in magnitude, so every number names its product.

Two Reanalyses, One Observational Check



A factor of 3.3 apart in mean peak SWE.
MERRA-2 exceeds ERA5's mean in 0 of 46 years.

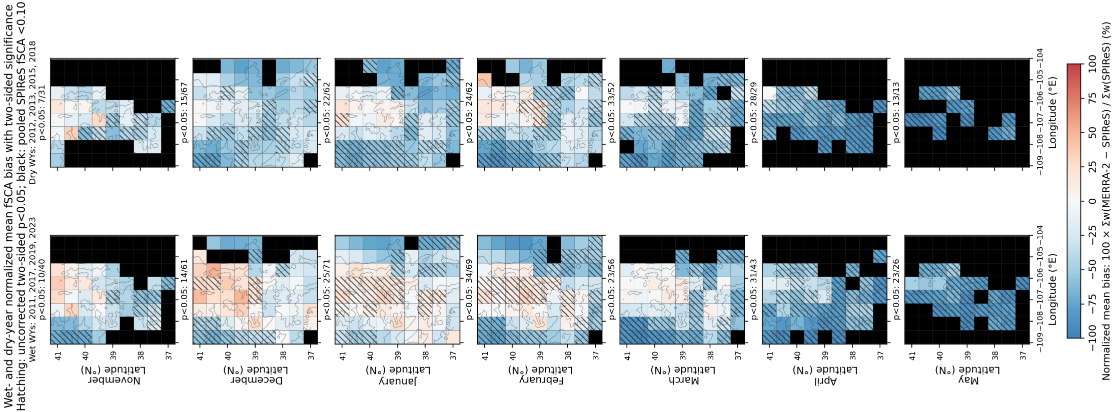
Yet they rank the years alike: $\rho = 0.73$



Each series scaled by its own peak.
SNOTEL arbitrates timing only.

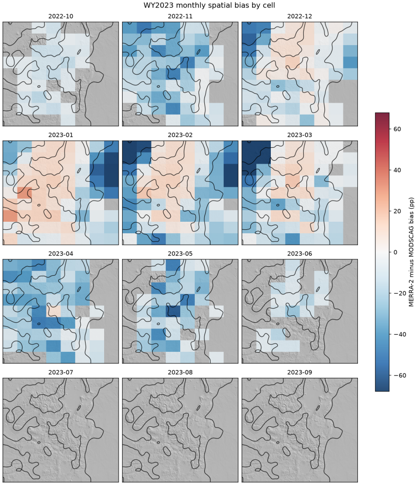
MERRA-2 peaks two months early

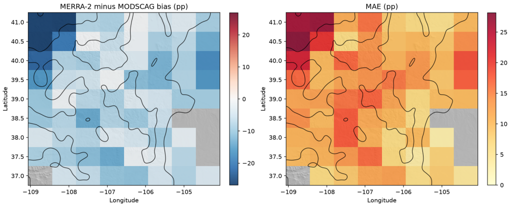
fSCA Bias, Wet Years against Dry

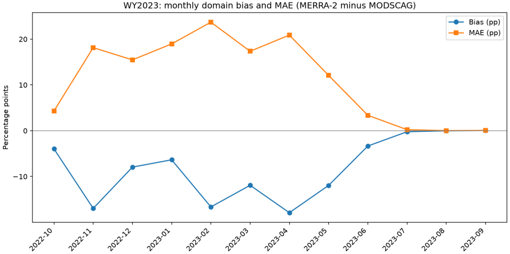


MERRA-2 minus SPIReS, normalized mean fSCA bias, WY2010–2023. Dry water years above, wet below; November at left, May at right. Hatching marks uncorrected two-sided $p < 0.05$.

fSCA Bias and MAE, Water Year 2023







MERRA-2 runs low against the satellite through accumulation and converges at melt-out.